
Page 10 of 10

File: Rsense_protection_power_output.kicad_sch

File: CC_CV_OVP.kicad_sch

 KK1
M3X8/D7985B

 KK2
M3X8/D7985B



KK5
SK96/84SA

© KK8
B3/BN2842

 KK7
M3X8/D7985B

KK6
SP400-0.007-00-104

KK9
PCB-MB-01

KK10
PCB-MB-01



<https://www.envox.eu>

Power module with SMPS pre-regulator, remote sense & programming
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)

Envox d.o.o.

Sheet: /

File: EEZ_DIB_DCP405plus.kicad_sch

Title: EEZ DIB DCP405+ power module

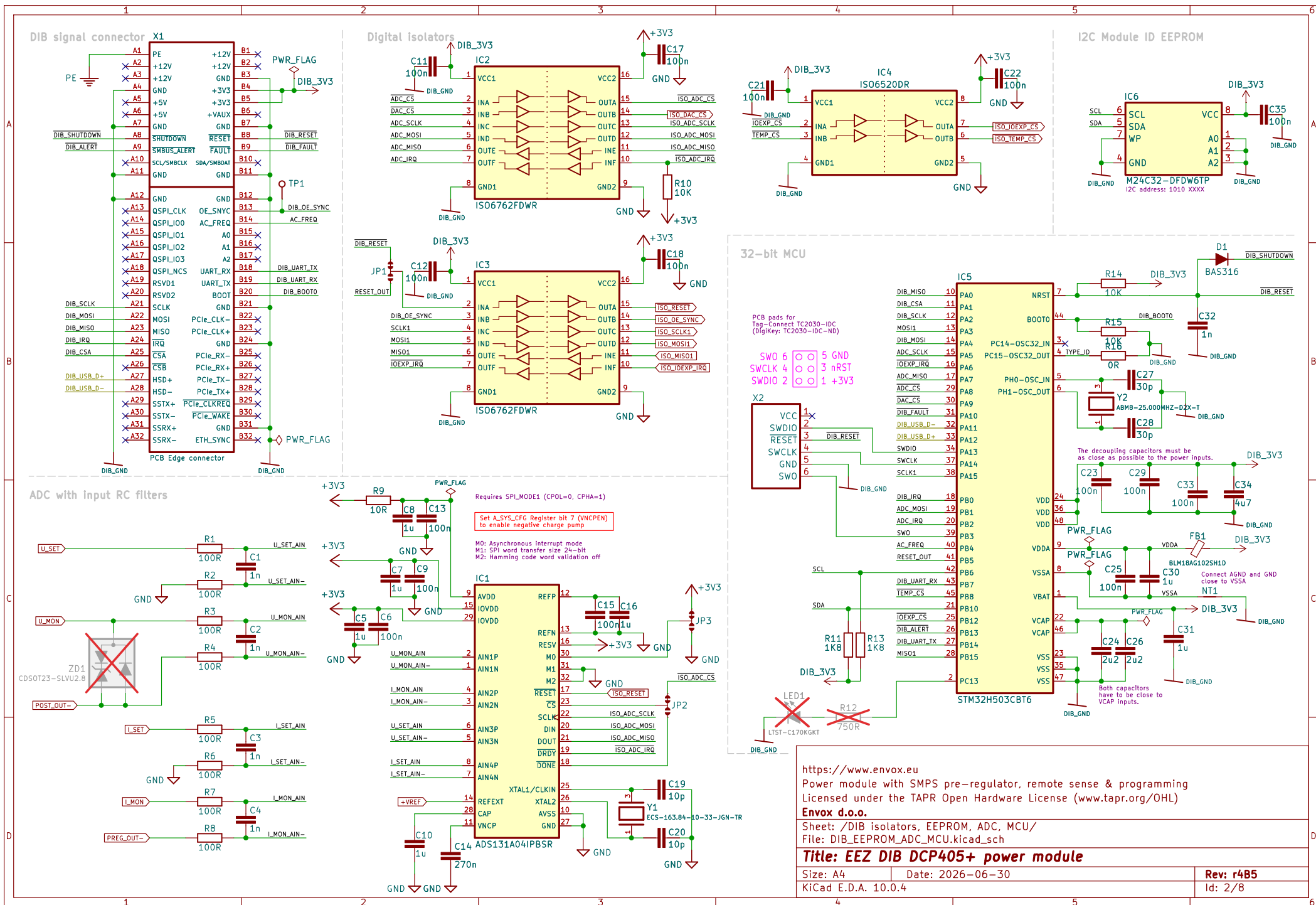
Size: A4

Date: 2026-06-30

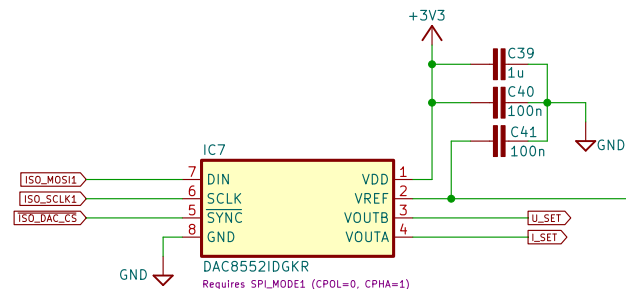
SIZE: A1	
KiCad E.D.A. 10.0.4	

Rev: r4B5

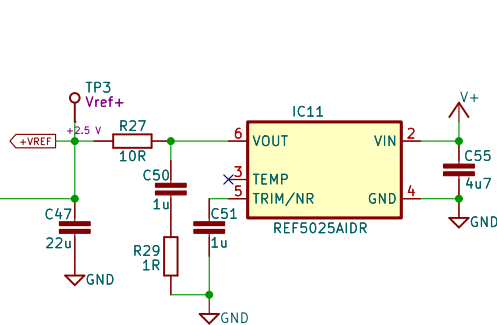
Id: 1/8



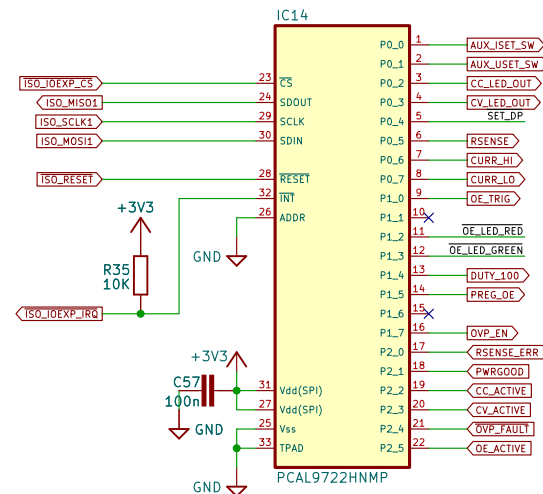
16-bit DAC (set voltage and current)



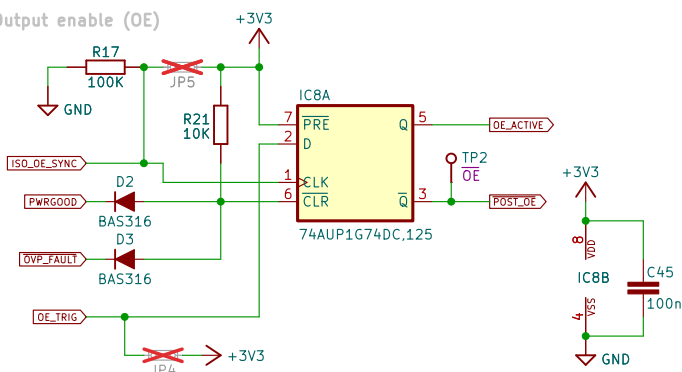
Voltage reference



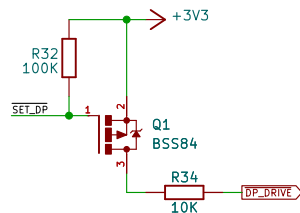
SPI 22-bit I/O Expander



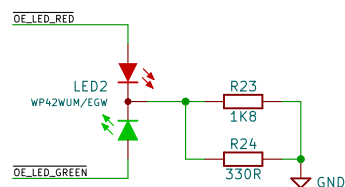
Output enable (OE)



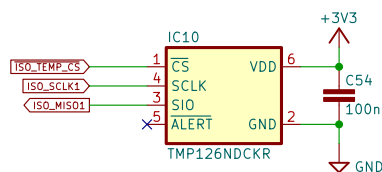
Down-programmer



OE Indicator



SPI Temperature sensor



<https://www.envox.eu>

Power module with SMPS pre-regulator, remote sense & programming
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)

Envox d.o.o.

Sheet: /DAC, OE, Vref, IOEXP, Temp/

File: DAC_Vref_IOEXP_Temp.kicad_sch

Title: EEZ DIB DCP405+ power module

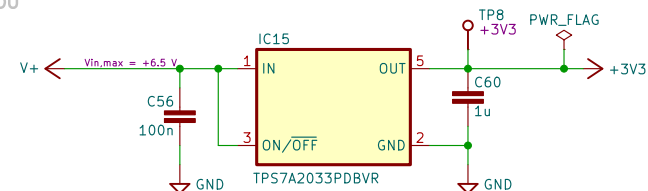
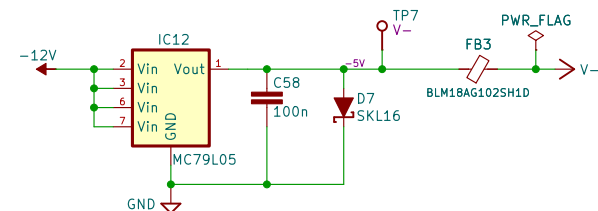
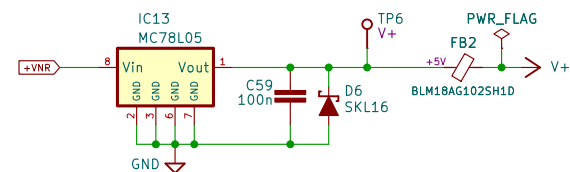
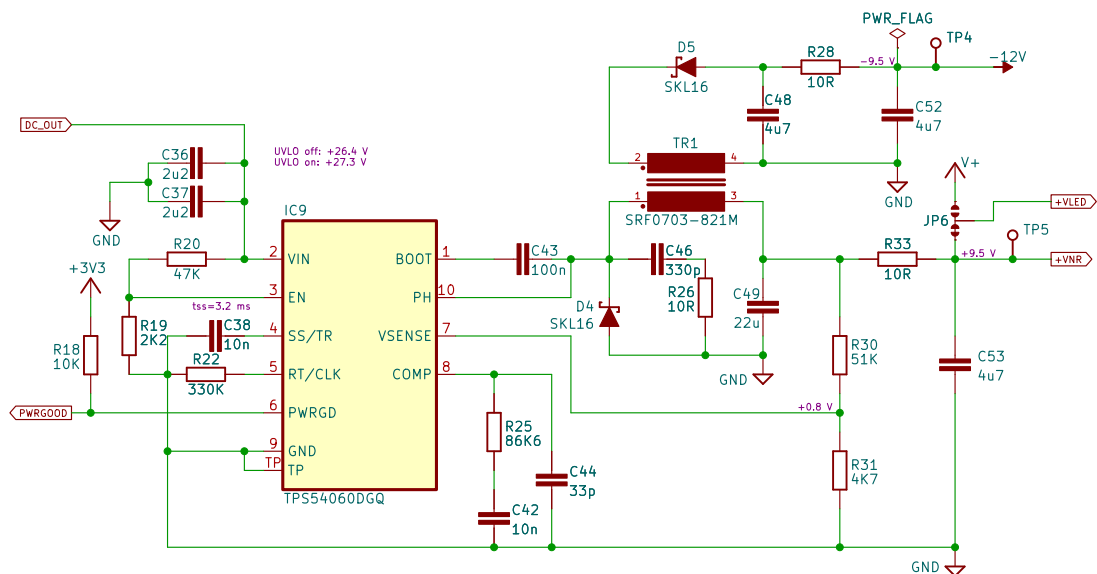
Size: A4

Date: 2026-06-30

Rev: r4B5

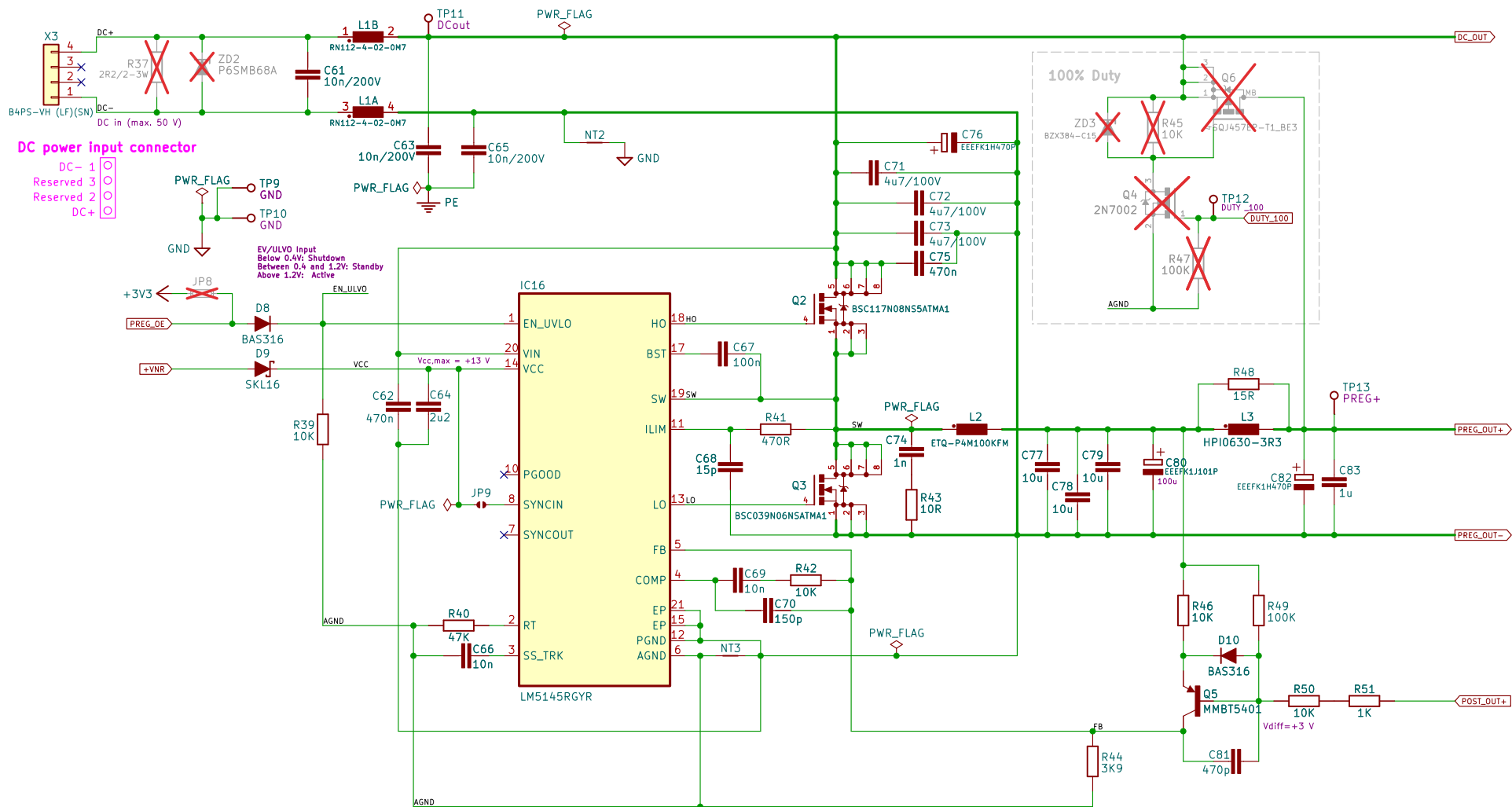
KiCad E.D.A. 10.0.4

Id: 3/8

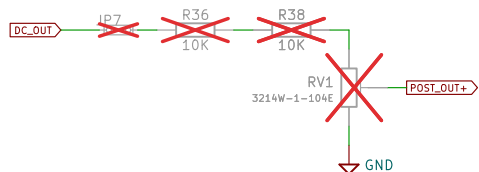


Id: 3/8

Low EMI power pre-regulator



Test point



<https://www.envox.eu>

Power module with SMPS pre-regulator, remote sense & programming
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)

Envox d.o.o.

Sheet: /Power pre-regulator/

File: Power_pre-regulator.kicad_sch

Title: EEZ DIB DCP405+ power module

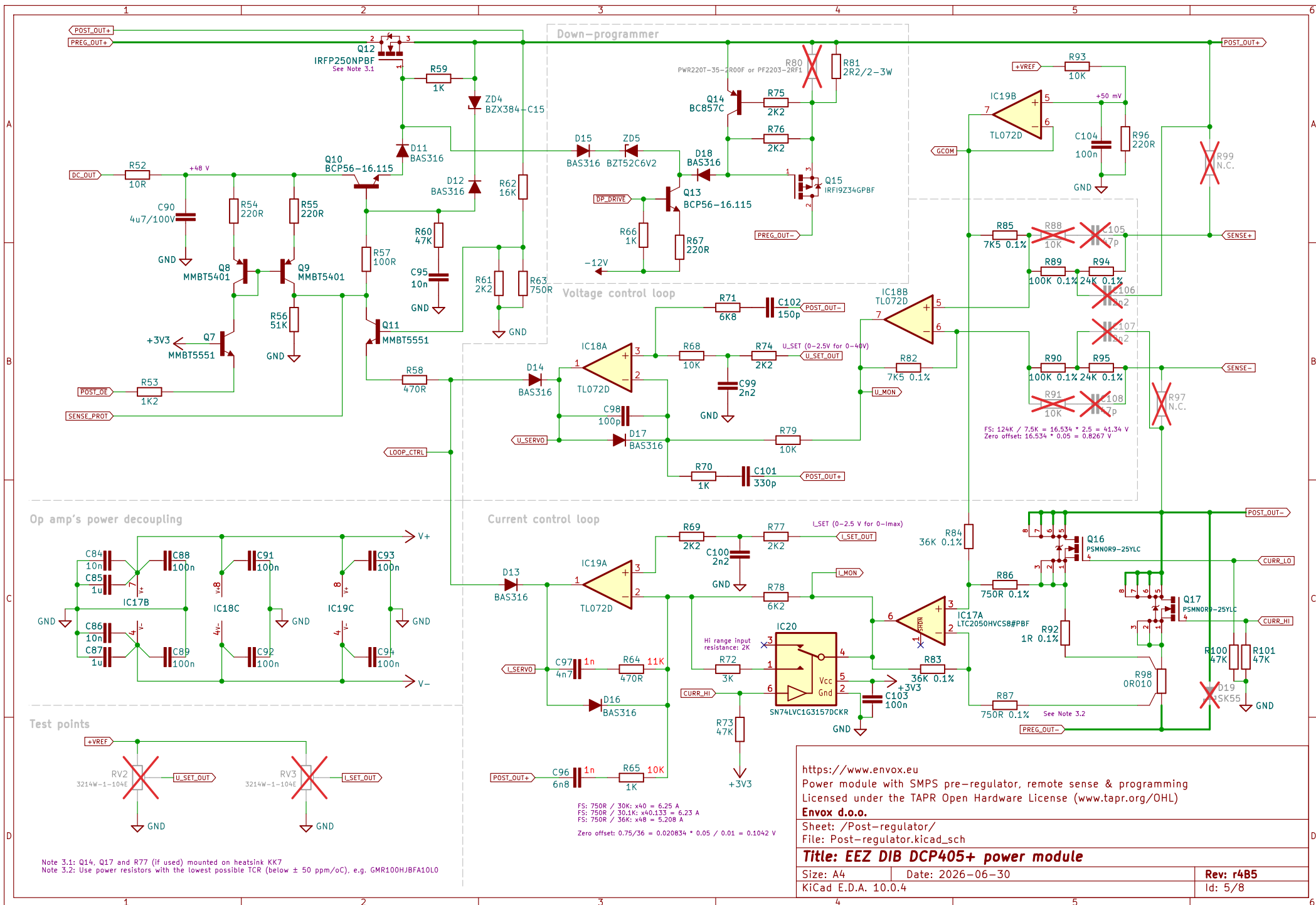
Size: A4

Date: 2026-06-30

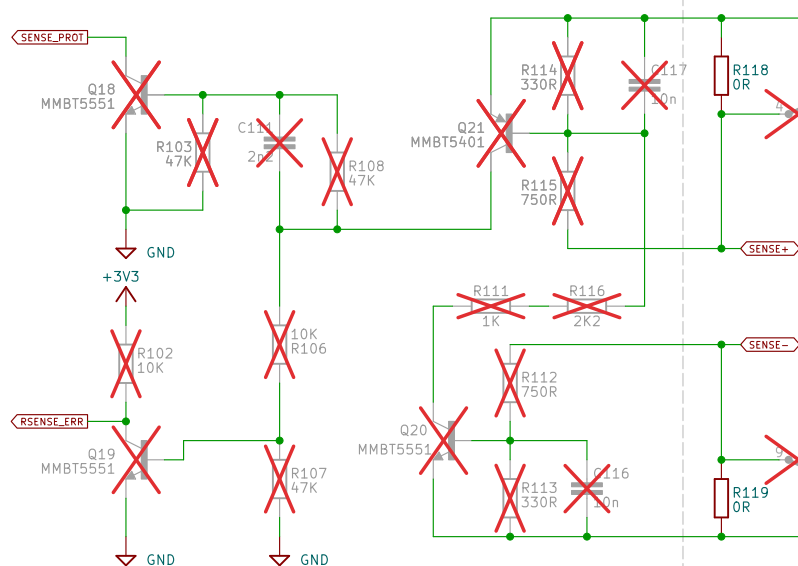
KiCad E.D.A. 10.0.4

Rev: r4B5

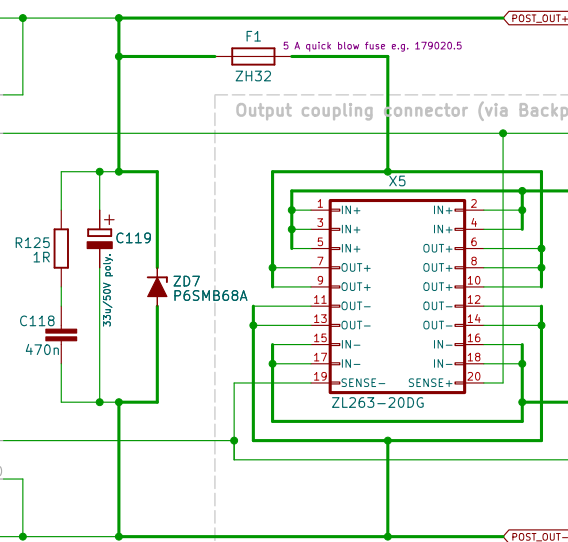
Id: 4/8



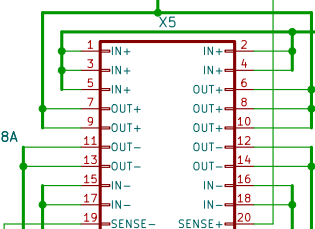
Remote sense reverse polarity detection



Power output section



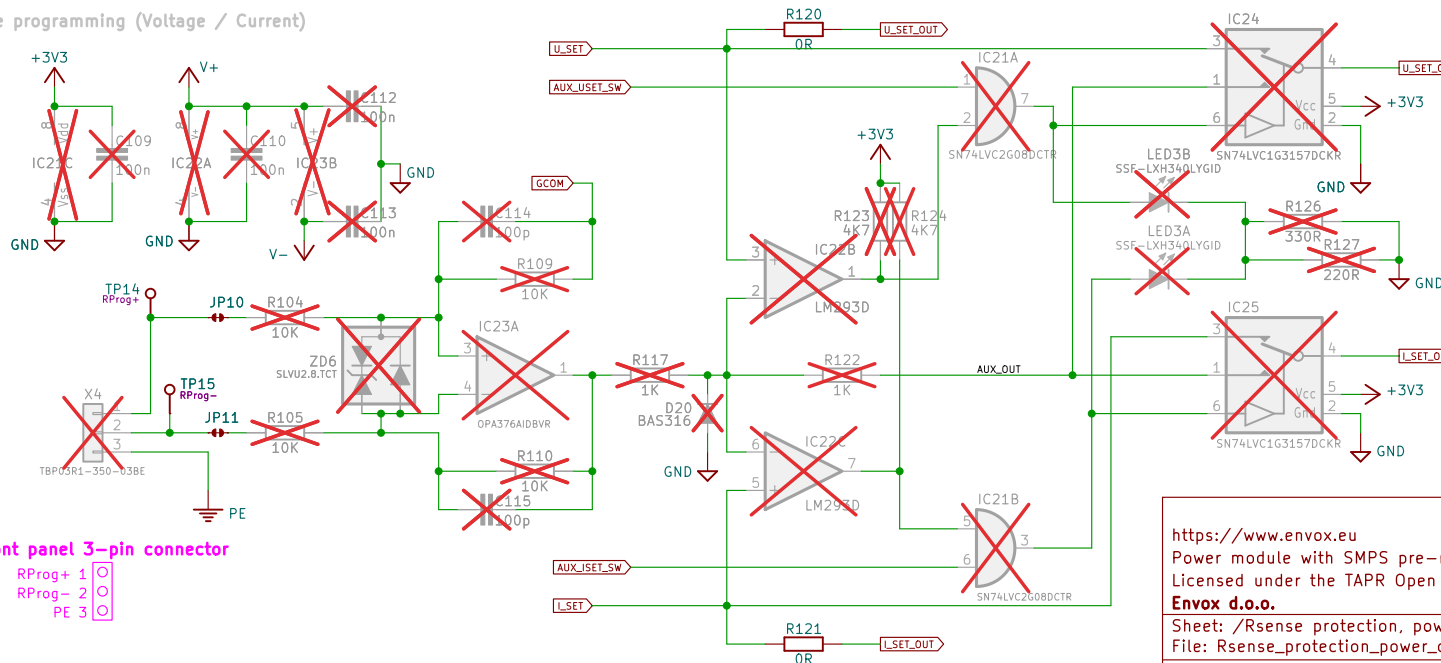
Output coupling connector (via Backplane PCB)



20-pin DIB 2.0 power socket

OUTC+ 1	2 OUTC+
OUTC+ 3	4 OUTC+
OUTC+ 5	6 POST_OUT+
POST_OUT+ 7	8 POST_OUT+
POST_OUT+ 9	10 POST_OUT+
POST_OUT- 11	12 POST_OUT-
POST_OUT- 13	14 POST_OUT-
OUTC- 15	16 OUTC-
OUTC- 17	18 OUTC-
SENSE- 19	20 SENSE+

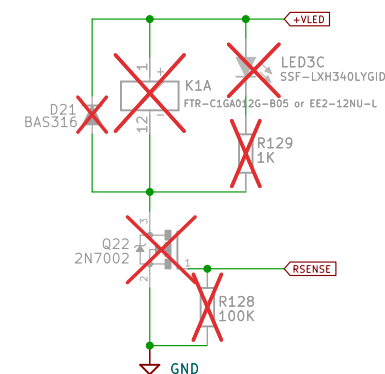
Remote programming (Voltage / Current)



Front panel 3-pin connector

RProg+ 1
RProg- 2
PE 3

Remote sense selection



<https://www.envox.eu>

Power module with SMPS pre-regulator, remote sense & programming
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)

Envox d.o.o.

Sheet: /Rsense protection, power output, Rprog/

File: Rsense_protection_power_output.kicad_sch

Title: EEZ DIB DCP405+ power module

Size: A4

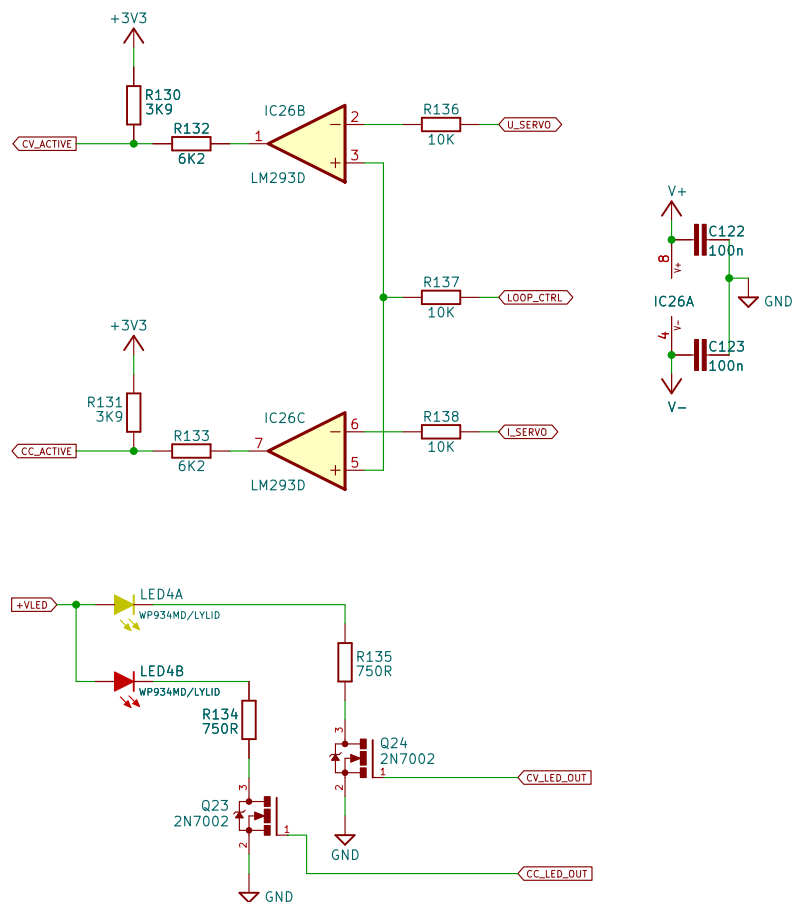
Date: 2026-06-30

KiCad E.D.A. 10.0.4

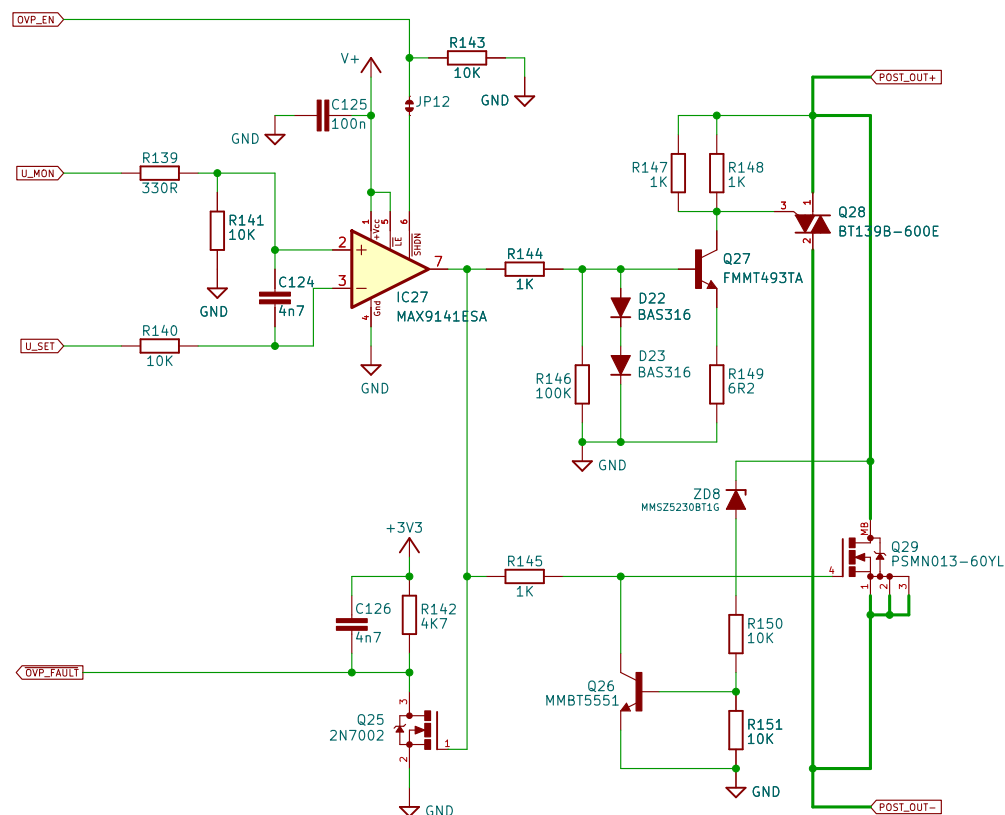
Rev: r4B5

Id: 6/8

Constant Voltage (CV) and Constant Current (CC) mode detection and indication



Over-voltage protection (OVP)



<https://www.envox.eu>

Power module with SMPS pre-regulator, remote sense & programming
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)

Envox d.o.o.

Sheet: /Mode detection, OVP/

File: CC_CV_OVP.kicad_sch

Title: EEZ DIB DCP405+ power module

Size: A4

Date: 2026-06-30

Rev: r4B5

KiCad E.D.A. 10.0.4

Id: 7/8